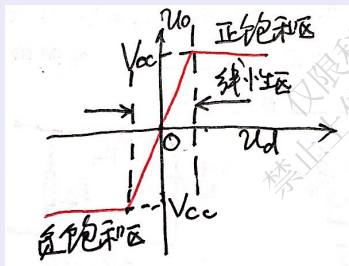


运算放大器 (Operational Amplifier)

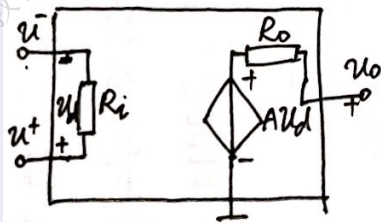
运算放大器功能

一种可以对输入信号进行求和、放大、积分、微分等运算功能的电路

OP的输入输出特性

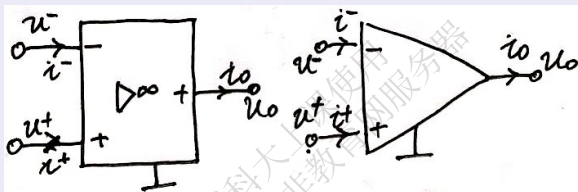


线性区工作模型



理想运放模型

理想运放的符号

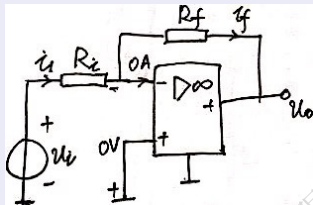


理想运放的端口特性

- 1 Virtual Open Circuit: $R_i = \infty$, $i^- = 0$, $i^+ = 0$
- 2 Virtual Short Circuit: $A = \infty$, $u_d = u^+ - u^- = 0$, $u^+ = u^-$

运算放大器电路的分析

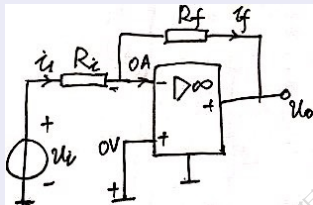
反向运算放大器电路



- 1 虚断, $i_1 = i_f$
- 2 虚短: $i_1 = u_i / R_i, i_f = -u_o / R_f$
- 3 $u_o = -R_f / R_i u_i$

运算放大器电路的分析

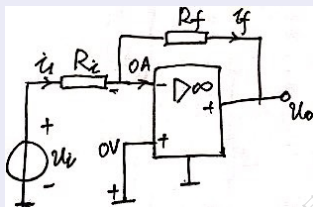
反向运算放大器电路



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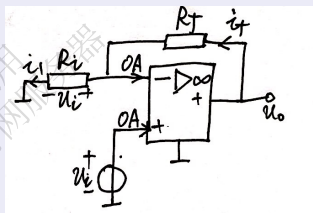
运算放大器电路的分析

反向运算放大器电路



- 1 虚断, $i_1 = i_f$
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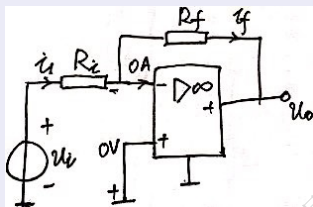
同相放大器



- 1 虚断: $i_f = i_1$
- 2 虚短: $u_o = R_f i_f + u_i, i_1 = u_i/R_i$
- 3 $u_o = (1 + R_f/R_i) u_i$

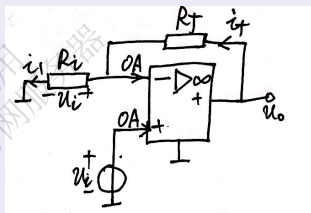
运算放大器电路的分析

反向运算放大器电路



- 1 虚断, $i_1 = i_f$
- 2 虚短: $i_1 = u_i/R_i, i_f = -u_o/R_f$
- 3 $u_o = -R_f/R_i u_i$

同相放大器



- 1 虚断: $i_f = i_1$
- 2 虚短: $u_o = R_f i_f + u_i, i_1 = u_i/R_i$
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